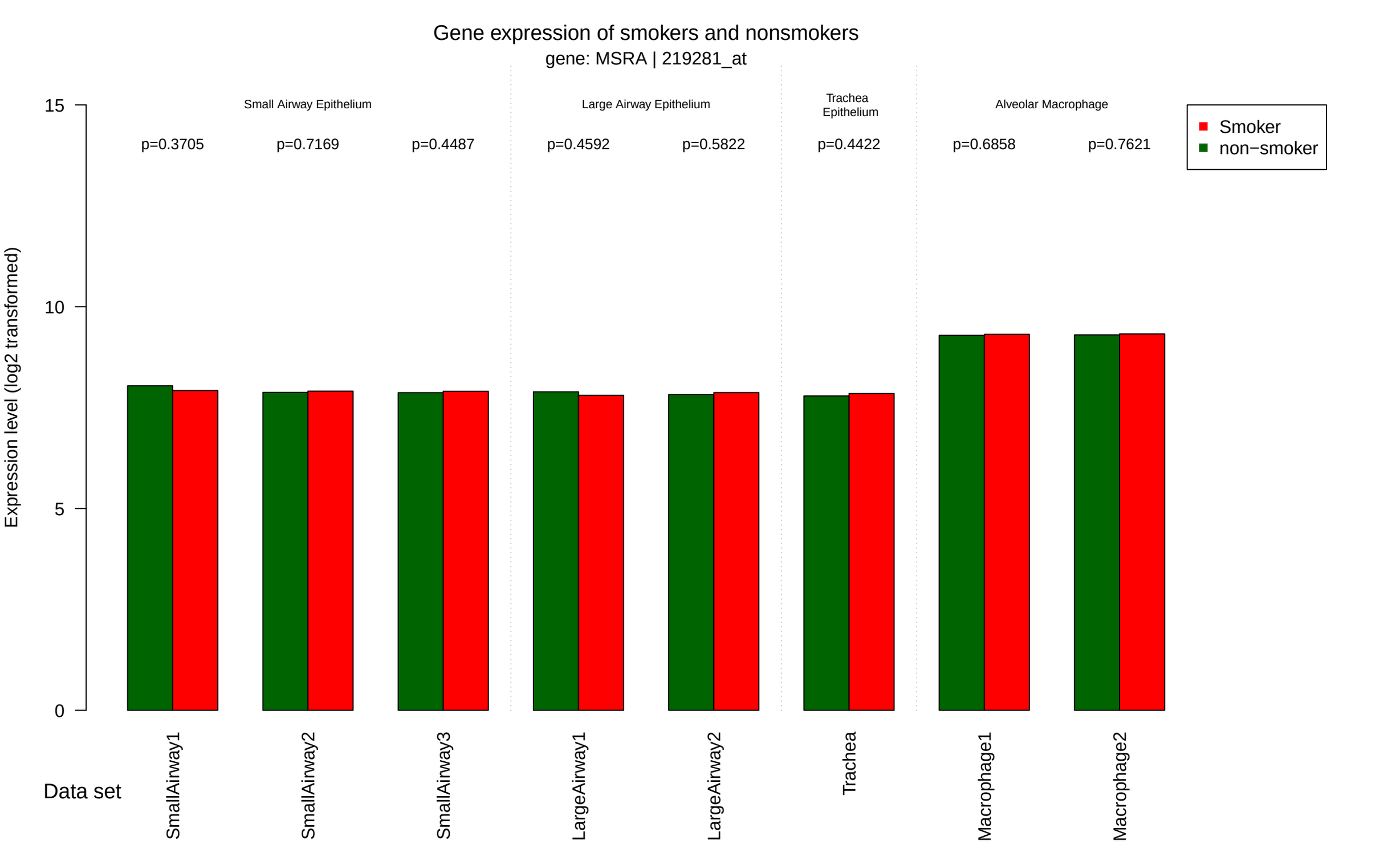
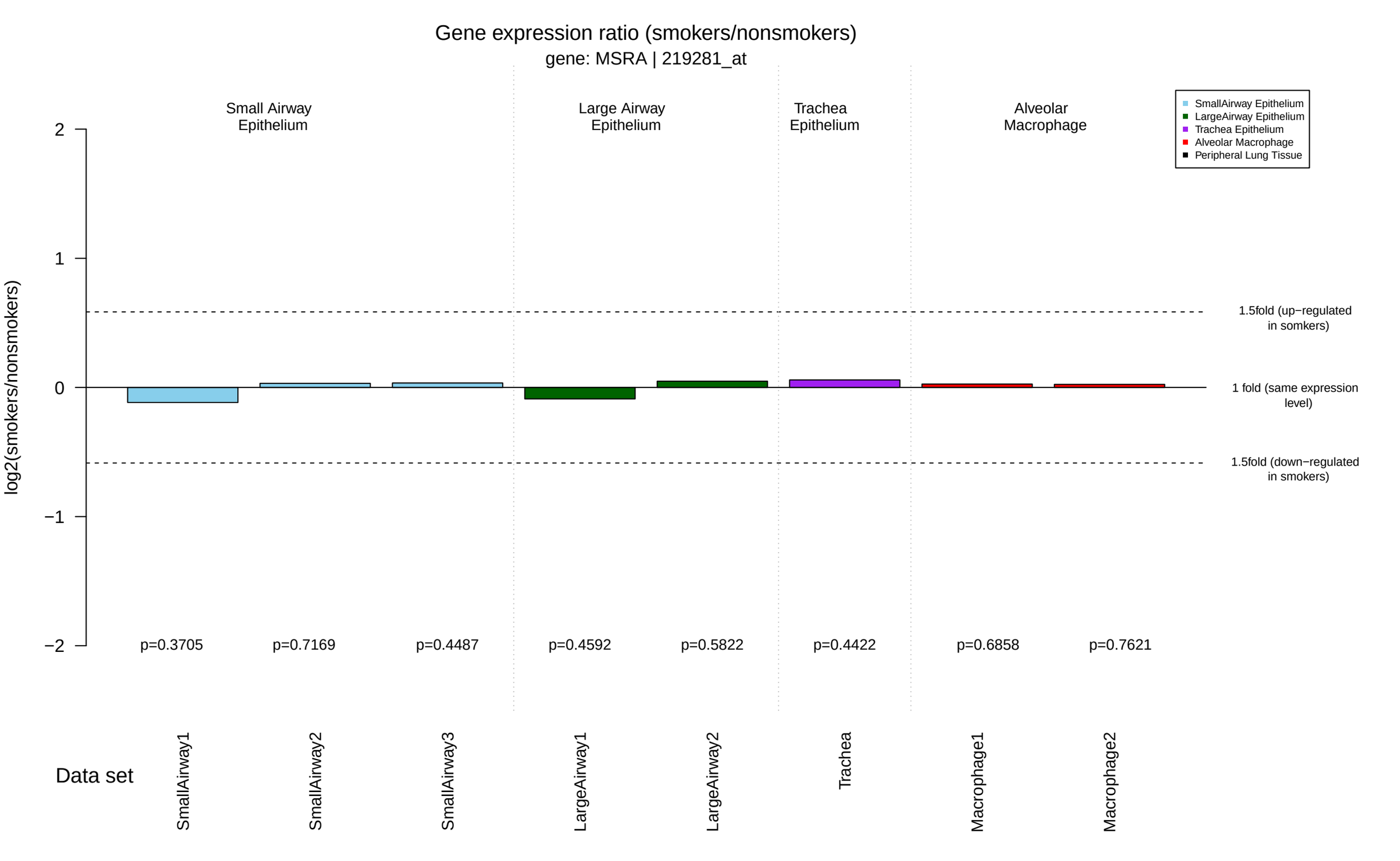
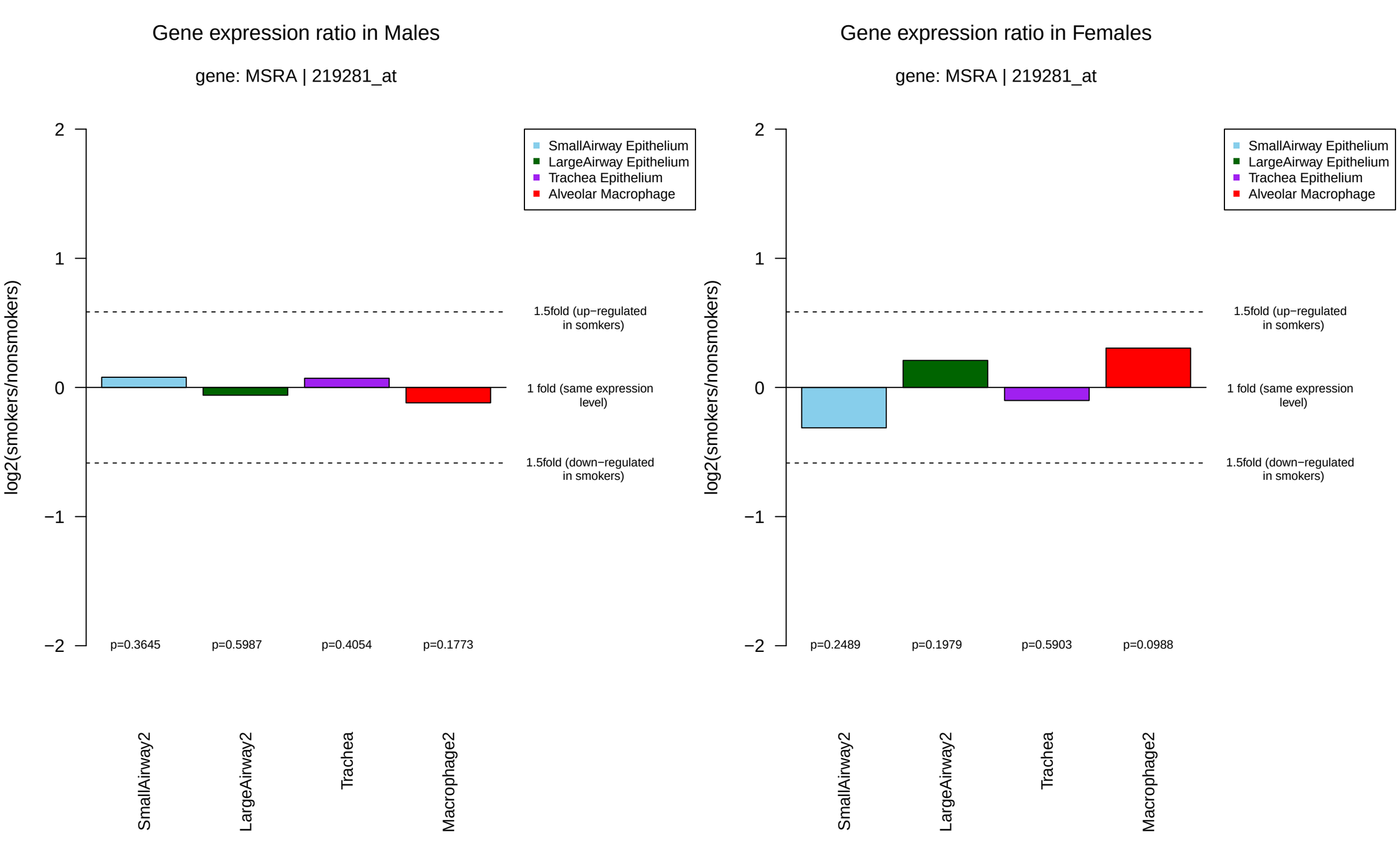
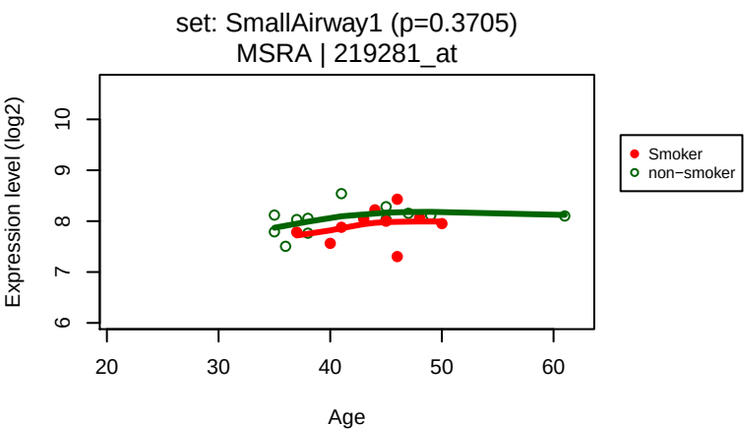
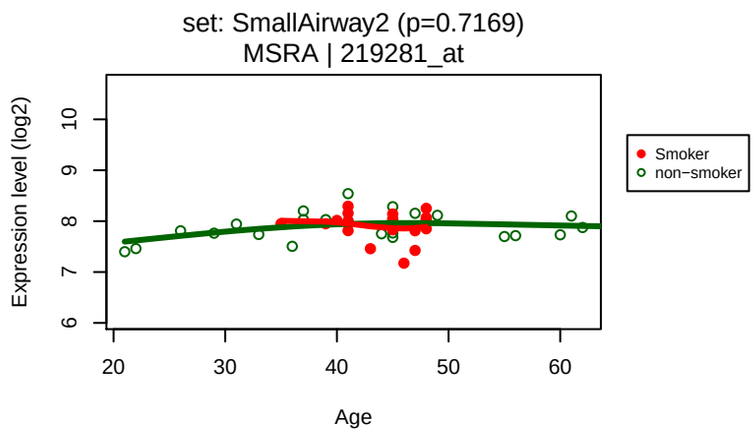
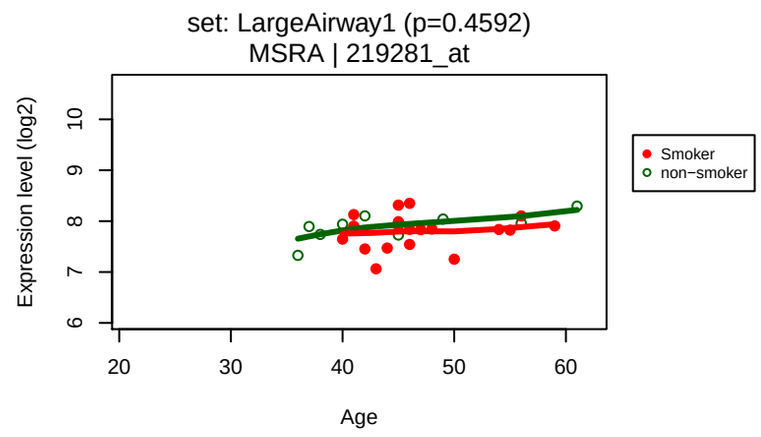

Gene expression of smokers and nonsmokers
gene: MSRA | 219281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3705
p=0.7169
p=0.4487
p=0.4592
p=0.5822
p=0.4422
p=0.6858
p=0.7621
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

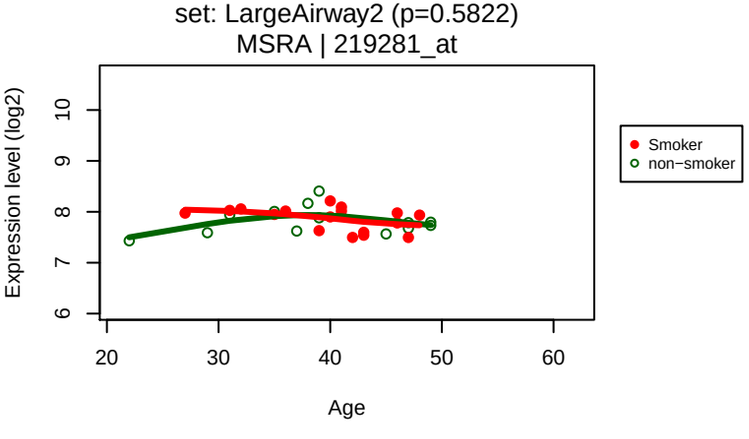

Gene expression ratio (smokers/nonsmokers)
gene: MSRA | 219281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3705
p=0.7169
p=0.4487
p=0.4592
p=0.5822
p=0.4422
p=0.6858
p=0.7621
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

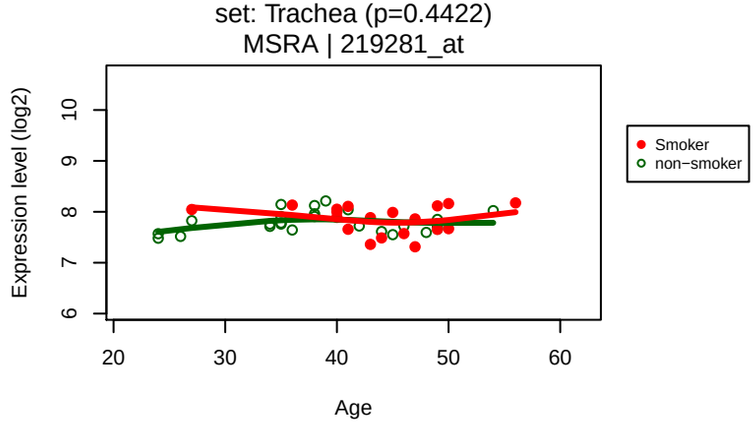

Gene expression ratio in Males
gene: MSRA | 219281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3645
p=0.5987
p=0.4054
p=0.1773
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MSRA | 219281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2489
p=0.1979
p=0.5903
p=0.0988
SmallAirway2
LargeAirway2
Trachea
Macrophage2

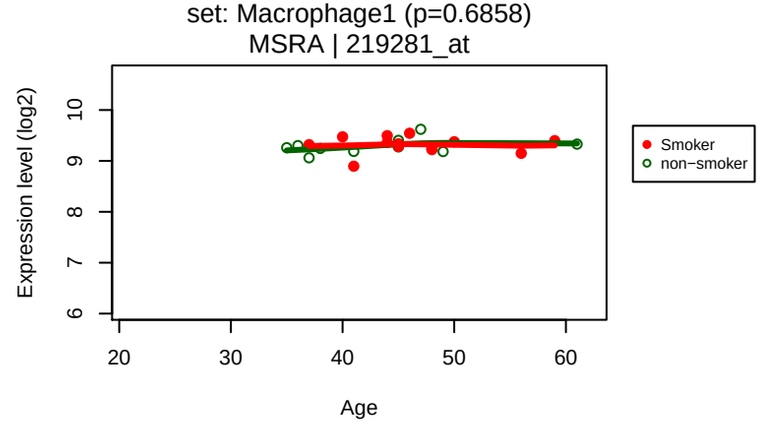

set: SmallAirway1 (p=0.3705)
MSRA | 219281_at
Expression level (log2)
Age
Smoker
non−smoker

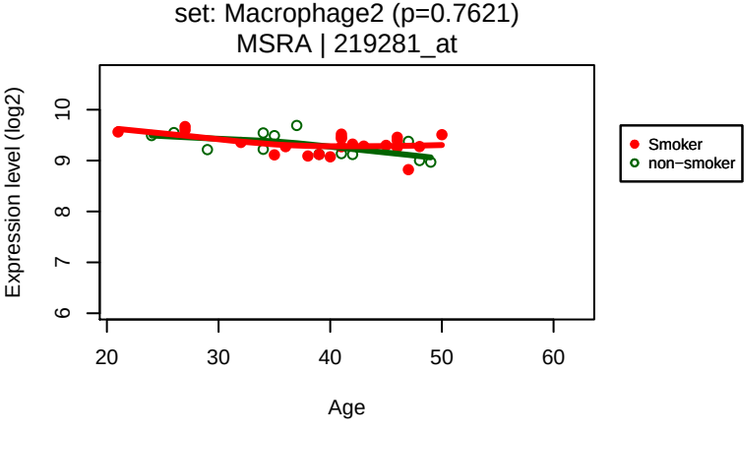

set: SmallAirway2 (p=0.7169)
MSRA | 219281_at
Expression level (log2)
Age
Smoker
non−smoker

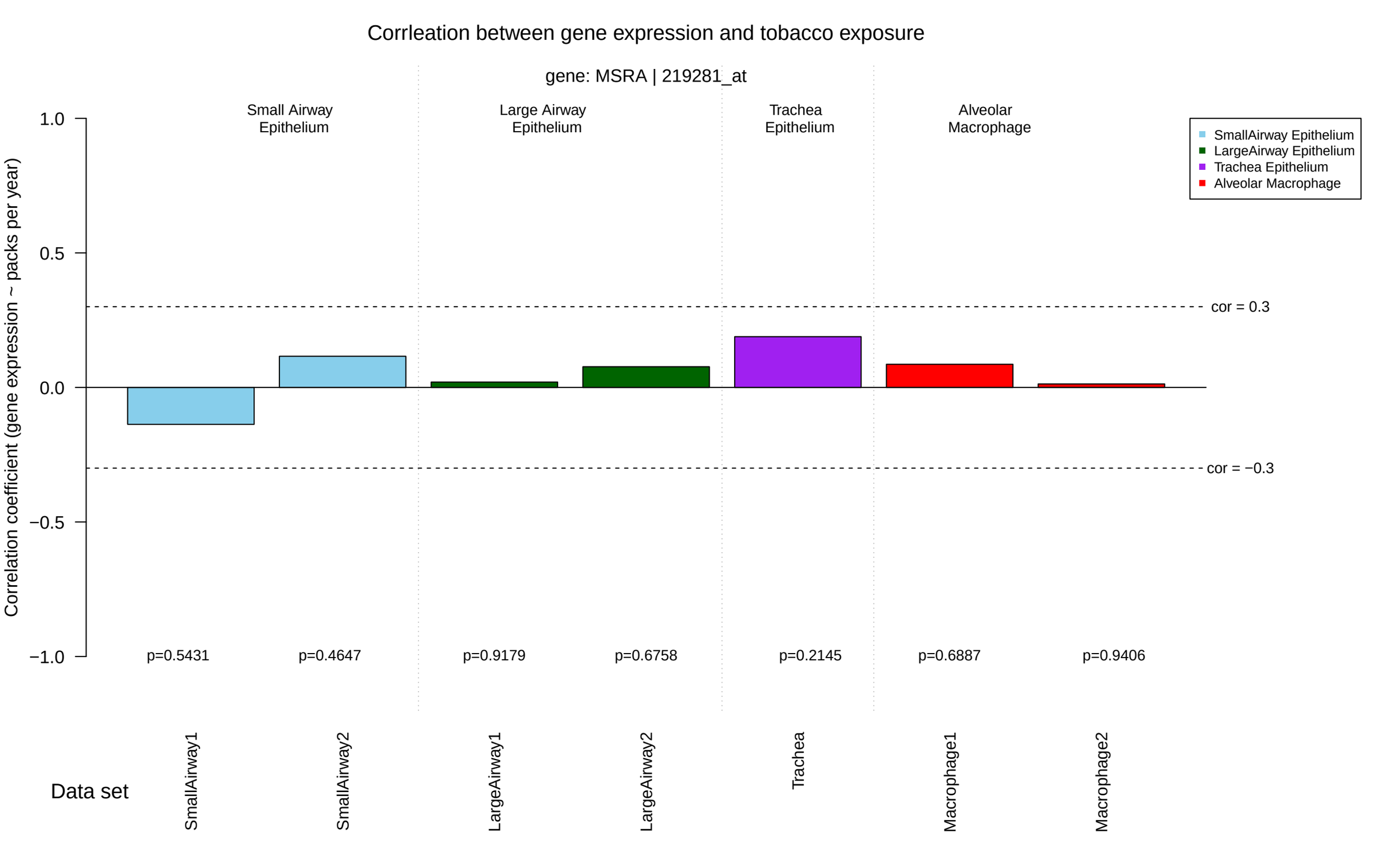

set: LargeAirway1 (p=0.4592)
MSRA | 219281_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5822)
MSRA | 219281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4422)
MSRA | 219281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6858)
MSRA | 219281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7621)
MSRA | 219281_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MSRA | 219281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5431
p=0.4647
p=0.9179
p=0.6758
p=0.2145
p=0.6887
p=0.9406
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2